EXAMPLE: T_6 JACOBIAN.

FIND ${}^{T_6}J_{11}$ AND ${}^{T_6}J_{41}$ FOR A REVOLUTE JOINT WHERE

$$ {}^{1}T_6 = A_1 A_2 A_3 A_4 A_5 A_6 = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ = \begin{bmatrix} C_1 (C_{234} C_5 C_6 - S_{234} S_6) - S_1 S_5 C_6 & C_1 (-C_{234} C_5 C_6 - S_{234} C_6) + S_1 S_5 S_6 & C_1 (C_{234} S_5) + S_1 C_5 & C_1 (C_{234} a_4 + C_{23} a_3 + C_2 a_2) \\ S_1 (C_{234} C_5 C_6 - S_{234} S_6) + C_1 S_5 C_6 & S_1 (-C_{234} C_5 C_6 - S_{234} C_6) - C_1 S_5 S_6 & S_1 (C_{234} S_5) - C_1 C_5 & S_1 (C_{234} a_4 + C_{23} a_3 + C_2 a_2) \\ S_{234} C_5 C_6 + C_{234} S_6 & -S_{234} C_5 C_6 + C_{234} C_6 & S_{234} S_5 & S_{234} a_4 + S_{23} a_3 + S_2 a_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$^{T_6}J_{11} = \left(-n_x \cdot p_y + n_y p_x\right)$$

$$= -\left[C_1\left(C_{234} C_5 C_6 - S_{234} S_6\right) - S_1 S_5 C_6\right] \times$$

$$\left[S_1\left(C_{234} a_4 + C_{23} a_3 + C_2 a_2\right)\right]$$

$$+ \left[S_1\left(C_{234} \cdot C_5 \cdot C_6 - S_{234} \cdot S_6\right) + C_1 S_5 C_6\right] \times$$

$$\left[C_1\left(C_{234} a_4 + C_{23} a_3 + C_2 a_3\right)\right]$$

$$= S_5 C_6 \left(C_{234} a_4 + C_{23} a_3 + C_2 a_2\right)$$

$$^{T_6}J_{41} = n_z = S_{234} C_5 C_6 + C_{234} S_6.$$